Ans 1a)

If a solution is written with Or, it means both are acceptable.

i. git clone https://github.com/ShoeMart/ShopEase.git
ii. git config user.name "ShopOps"
git config user.email ops@shoemart.com
Or
git config --global user.name "ShopOps"
git config --global user.email ops@shoemart.com
iii. git branch hotfix-payment
git branch feature-discounts
iv. git checkout hotfix-payment
git add payment-fix.js
git commit -m "Fix payment rounding bug"
v. git checkout feature-discounts
git add coupons.md
git commit -m "Document coupon rules"
vi. git rm coupons.md
git commit -m "Remove coupons draft"

Or
- Manually delete coupons.md (Writing this line is not mandatory)
git add coupons.md
git commit -m "Remove coupons draft"
vii. git reset --hard f00d1123
viii. git checkout master
git merge hotfix-payment
git merge feature-discounts
git push / git push origin master
(The second and third line can be interchanged, also both git push command can be used as the fourth line)

Marking Rubrics: 6 marks
i. 0.5
ii. 0.5
iii. 0.5
iv. 1
v. 1
vi. 1
vii. 0.5
viii. 1

Ans 2a)

Requirements specification is the process of documenting the requirements identified in the analysis step in a clear, consistent, and unambiguous manner.
The models used in this step are: ER diagrams, data flow diagrams (DFDs), function decomposition diagrams (FDDs), data dictionaries, etc. (may pick any two)

Marking Rubrics: 2 marks
What is Requirements Specification – 1
Mention two models – 1

Ans 2b)

Case 1 - Attack: Ransomware/Malware (Ransomware will be more specific)
Reasons: In a ransomware attack, malware encrypts important files or systems and displays a ransom note demanding payment (often in cryptocurrency) to restore access. In the above scenario, the LMS was shut down, all course materials and submissions were encrypted, and a bitcoin ransom note was shown. Therefore, this matches the pattern of a Ransomware Attack.

Case 2 - Attack: Password brute-force
Reasons: In a brute-force attack, an attacker repeatedly tries multiple username-password combinations until the correct one is found, often from unusual or foreign IP addresses. In the above scenario, multiple log-in attempts from foreign IPs were detected, and accounts were hijacked with fake announcements and assignments. Therefore, this matches the pattern of a Password Brute-Force Attack.

Marking Rubrics: 4 marks
Case 1 - 2 marks
o Correct Attack – 1
o Proper Reasoning – 1

Case 2 – 2 marks
o Correct Attack – 1
o Proper Reasoning – 1

Ans 3a)

The first is architectural documentation, and the second is marketing documentation.

Rubric
1 - correct identification
1 - proper reasoning (relate to scenario)

Ans 3b)

Spiral Model
Reasoning:
1. The project is highly critical for national elections, so risk management is essential.
2. System must work in rural areas with uncertain internet connectivity, which adds technical risks.
3. Different biometric devices and software solutions need to be tested in small pilot projects before final deployment. The Spiral Model allows iterative development with prototypes, which fits the pilot-testing requirement.
4. Officials require formal evaluations and approvals after each stage, which aligns with the review phase in every spiral loop

Rubric
1 - correct model
1 - model description
2 - proper reasoning (relate to scenario)

Ans 4a)

1. Daily Scrum is every day for about 15 minutes.
Sprint Planning is once at the start of the sprint and is longer.
2. Daily Scrum checks progress and plans the next day.
Sprint Planning sets the sprint goal and chooses the work for the whole sprint.

Rubric
1 for each difference

Ans 4b)

1. Shila sits with the team, answers on the spot, and accepts changes : On-Site Customer
2. Write a small check that fails first, then just enough code to pass : Test-Driven Development (TDD)
3. Clean up names and split big functions without changing behavior : Refactoring
4. Merge every change to main; automatic run checks everything; fix breaks immediately : Continuous Integration (CI)

Rubric
1 for each identification and reasoning

Ans 5a)

Multiple subclasses implement a method with similar or identical behavior, causing code duplication. To refactor such codes, we need to pull up the methods. Here, we move the common method from the subclasses to the superclass. Therefore, subclasses can inherit the methods instead of redefining them.

Rubrics:
- 1 mark for problem identification.
- 1 mark for the solution idea.

Ans 5b)

```java
public class Sport {
    private String name;
    private int players;

    public Sport(String name, int players) {
        this.name = name;
        this.players = players;
        System.out.println("Sport: " + name + " is played among " + players + "
        ↪ players");
        printTeamSizeCategory();
    }

    private void printTeamSizeCategory() {
        if (players > 10) System.out.println("Big team game");
        else if (players > 5) System.out.println("Medium team game");
        else if (players > 0) System.out.println("Small team game");
    }
}
```

```java
public class Tournament {
    private String name;
    private int teams;

    public Tournament(String name, int teams) {
        this.name = name;
        this.teams = teams;
        System.out.println("Tournament: " + name + " is created with " + teams + "
        ↪ teams.");
    }

    public void printTotalMatches() {
        int totalMatches = (teams * (teams - 1))/2;
        System.out.println("Total number of matches: " + totalMatches);
    }
}
```

Rubrics:

- 2 marks for properly renaming all the variables.
- 1 mark for encapsulating and using access modifier.
- 1 mark for using algorithm substitution.